

MAHENDRA PRATAP SINGH

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PROFESSIONAL SUMMARY

- Over **22 years** of experience in **Software Testing, Test Automation, Quality Engineering, and Test Management** across Banking, Financial Services, Health Insurance, Government, Land Information, and Telecom.
 - **16+ years** of hands-on experience in **Test Automation Architecture**, designing scalable **Web, API, Mobile**, and backend automation solutions with a strong focus on **shift-left testing** and continuous **quality engineering**.
 - Proven experience leveraging **Generative AI, GitHub Copilot, and AI Agents** across **test planning, test design, framework creation, code generation, debugging, and automation maintenance**, including development of reusable **AI-assisted skills** for generating **Test Plans, Test Requirement Traceability Matrices (RTM), Playwright/REST Assured frameworks, and automated tests** from requirements and approved test plans.
 - **10+ years** of **test management** and technical leadership experience, including leading cross-functional teams of 20+ professionals across multiple geographies.
 - Strong expertise in **Test Strategy, Test Planning, Risk-Based Testing, Requirement Traceability, Defect Management, Quality Governance, and Agile/SAFe** delivery.
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TECHNICAL SKILLS

- **AI-Assisted Quality Engineering:** Generative AI, AI Agents, GitHub Copilot, Prompt Engineering, AI-Assisted Test Planning & Test Design, Test Plan & RTM Generation, AI-Assisted Test Automation & Framework Generation, Automated Test Code Generation.
- **Web Application Testing:** Playwright, Cypress, React Testing Library, Selenium WebDriver, Browser Use, Selenium Basic.
- **API Testing:** REST Assured, Postman, Insomnia, Spring Boot, OpenAPI, Swagger Supertest, SoapUI, Karate.
- **Mobile Testing:** XCUITest, Espresso, Appium, Experitest SeeTest, Appium Studio.
- **Desktop Application:** UFT, QTP, WinRunner.
- **Automation Architecture & Frameworks:** Page Object Model (POM), Behaviour-Driven (BDD), TestNG, JUnit, Jest, Mocha, Cucumber, Concordion.
- **Programming Languages:** TypeScript, JavaScript, Java, Kotlin, Python, Swift, C#, VB.NET, C/C++, VBScript, TSL.
- **CI/CD, DevOps & Build Tools:** Jenkins, GitHub Actions, Azure DevOps, Git, GitHub, GitLab, Bitbucket, Maven, Gradle, NodeJS, WinCVS, Tortoise SVN.
- **Cloud & Containerization:** AWS, Azure, Firebase, Kafka, Docker.
- **Project Management Tools:** Agile, SAFe, Scrum, Kanban, Waterfall, Jira, Confluence, Miro, InVision.
- **Defect Tracking:** ALM/Quality Center, TestDirector (TD), Change Synergy, SharePoint.

- **Databases:** IBM Db2, Informix, AWS DynamoDB, Oracle, SQL.
 - **Application Development & CMS Technologies:** React, Spring Framework, UIKit, SwiftUI, Combine Framework, Vignette CMS, Adobe Analytics.
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WORK EXPERIENCE

Senior Test Automation Engineer (Contract)

Bank of New Zealand

July 2025 – Present

- Leveraged Generative AI, GitHub Copilot, and AI Agents to accelerate test planning, test design, automation development, code generation, debugging, and maintenance.
- Developed reusable AI-assisted generate-test-plan and generate-pw-test skills for generating Test Plans, Test Requirement Traceability Matrices (RTM), automation frameworks, and automated tests from requirements and approved test plans.
- Designed and implemented scalable frontend and backend/API test automation frameworks using Playwright with TypeScript and REST Assured with Java, with reusable fixtures, utilities, and automation patterns.
- Integrated automated testing into CI/CD pipelines, enabling continuous testing, faster feedback, and early defect detection while applying shift-left testing principles.
- Collaborated with developers, business analysts, product owners, and stakeholders to define test scenarios and deliver comprehensive functional and integration test coverage.

Senior Test Automation Engineer (Contract)

Reserve Bank of New Zealand (RBNZ)

April 2025 – June 2025

- Leading the design and development of scalable API test automation frameworks, seamlessly integrating them into Azure DevOps CI/CD pipelines to support continuous testing and early defect detection.
- Defining and implementing a comprehensive test data management strategy, including data provisioning and masking, to ensure consistent, reusable, and secure test data across multiple test environments.

Senior Test Automation Engineer (Contract)

Land Information New Zealand (LINZ)

November 2022 – March 2025

- Developed and tested frontend applications using React, React Testing Library, and TypeScript, ensuring functionality and quality across user-facing features.
- Implemented and tested backend solutions using Spring Boot and Kotlin, leveraging Swagger and OpenAPI specifications for API design, documentation, and integration validation.
- Conducted end-to-end testing using Playwright and managed CI/CD pipelines and version control with Git, GitHub, GitHub Actions, Docker, and AWS, supporting reliable and efficient delivery.

- Applied SAFe Agile methodologies, using Jira, Confluence, Miro, and Figma to support planning, collaboration, and alignment across cross-functional teams.

Senior Test Automation Engineer (Contract)

New Zealand Customs Service

July 2022 - November 2022

- Led test automation, designing, developing, and maintaining automation frameworks and scripts for web applications (Selenium, Cucumber) and API testing (REST Assured), created Postman collections to support manual testing.
- Implemented a Java-based tool for bulk API data creation, achieving a 60% reduction in effort, and integrated automated testing into CI/CD pipelines using Jenkins and Git.
- Applied SAFe Agile methodologies using Jira and Confluence to support planning, collaboration, and alignment across cross-functional teams.

Senior Test Automation Engineer (Contract)

Bank of New Zealand Limited

December 2021 - July 2022

- Led API test automation, designing, developing, and maintaining automation frameworks and scripts using TypeScript, Jest, Supertest, AWS SDK, AWS DynamoDB, and Kafka.
- Utilised Swagger and OpenAPI specifications for API design and documentation, and developed Postman collections to support manual API testing.
- Integrated automated testing with CI/CD pipelines using Jenkins and Git, while applying SAFe Agile methodologies using Jira and Confluence.

Senior Test Automation Engineer (Contract)

Land Information New Zealand (LINZ)

April 2021 - December 2021

- Developed and tested frontend applications using React, React Testing Library, and TypeScript, ensuring functionality and quality across user-facing features.
- Implemented and tested backend solutions using Spring Boot and Kotlin, leveraging Swagger and OpenAPI specifications for API design, documentation, and integration validation.
- Conducted end-to-end testing using Playwright and managed CI/CD pipelines and version control with Git, GitHub, GitHub Actions, Docker, and AWS, supporting reliable and efficient delivery.
- Applied SAFe Agile methodologies, using Jira, Confluence, Miro, and Figma to support planning, collaboration, and alignment across cross-functional teams.

Senior Test Automation Engineer (Contract)

Bank of New Zealand Limited

August 2019 - April 2021

- Led test automation activities, designing, developing, and maintaining automation frameworks and scripts for mobile (XCUITest, Appium, SeeTest), web (Selenium, Concordian, Cypress), and API testing (Postman, REST Assured, Supertest, Jest).

- Integrated automated testing into CI/CD pipelines using Jenkins, Git, and Bitbucket, enabling continuous testing and faster feedback.
- Applied SAFe Agile methodologies using Jira, Confluence, Miro, and InVision to support planning, collaboration, and alignment across cross-functional teams.

Senior Test Automation Engineer (Contract)

Westpac New Zealand Limited

October 2017 - August 2019

- Led test automation activities, designing, developing, and maintaining automation frameworks and scripts for mobile (XCUITest, Appium, SeeTest), web (Selenium, Cucumber, Cypress), and API testing (Postman, REST Assured).
- Integrated automated testing into CI/CD pipelines using Jenkins, Git, and Bitbucket, enabling continuous testing and faster feedback.
- Applied SAFe Agile methodologies using Jira, Confluence, Miro, and InVision to support planning, collaboration, and alignment across cross-functional teams.

Manager

Capgemini Australia Pty Limited

June 2015 - October 2017

- Business & Technology Test Manager for ANZ Bank, leading testing across multiple concurrent Agile and Waterfall projects, including end-to-end test planning, execution, reporting, and quality assurance.
- Defined and executed robust test strategies covering functional, integration, regression, and performance testing, coordinating cross-functional teams to achieve successful project outcomes.
- Led end-to-end testing for implementation of the UPI China card payment scheme, validating integration across issuer/acquirer systems, payment gateways, and core banking systems, including EMV chip, contactless, and e-commerce transaction flows.
- Oversaw validation of the complete payment transaction lifecycle, including authorisation, clearing, settlement, chargebacks, and disputes, managed defect triage, test metrics, and go-live readiness sign-off with key stakeholders.
- Developed a web services data extraction and comparison utility, achieving a 30% reduction in testing effort, also created deployment plans and performed rigorous post-implementation verification to ensure successful integration and system performance.
- Managed client relationships and a team of 20, ensuring timely delivery and high client satisfaction while conducting performance appraisals, identifying training needs, setting goals, and developing high-performing teams.

Project Manager

Tech Mahindra Ltd

June 2014 - June 2015

- Developed and implemented mobile test automation strategies, reducing testing time by 50% and increasing test coverage by 40%.

- Led multiple client engagements and projects, managing resources and delivery to consistently meet timelines, quality standards, and client expectations.
- Headed the Mobile Application Test Automation Competency team, establishing automation standards and driving best practices across projects.
- Designed and implemented custom mobile automation solutions for RFPs/RFIs, supported by feasibility analysis, cost-benefit and ROI assessments, and POCs to enable data-driven decisions.
- Led recruitment and managed a team of 15, conducting performance appraisals, setting strategic goals, identifying training needs, and building high-performing teams focused on continuous improvement.

United Health Group, Gurgaon, India

IT Quality Consultant (Project Lead)

January 2010 - June 2014

May 2008 - September 2009

- Led feasibility analysis, automation solution design, and POCs, aligning automation strategies with project objectives and business needs.
- Saved \$0.5 million by developing HIPAA EDI file-to-database comparison utilities, improving data integrity, reducing manual effort, and supporting compliance with healthcare industry standards.
- Conducted automation effort estimation, cost-benefit analysis, and ROI assessments, optimising resource allocation and driving automation adoption across multiple projects through stakeholder demos and presentations.
- Designed and built a three-tier automation framework covering scripting, test data management, and execution, enabling scalable automation across multiple environments.
- Led and managed teams of 20+ in a matrix environment, driving cross-functional collaboration, recruitment, performance appraisals, strategic goal setting, training, and continuous team development.

Sapient Corporation, Gurgaon, India

Senior Associate QA L1

September 2009 - January 2010

June 2005 - April 2008

- Automated report generation, reducing effort by 75%, minimizing manual tasks, and enhancing accuracy by lowering error rates.
- Designed and implemented a QTP-based keyword-driven automation framework for web applications, increasing test efficiency and accuracy.
- Presented the automation framework to clients, highlighting its capabilities and benefits, and successfully integrated it across multiple projects.
- Mentored and managed team members, providing guidance, support, and fostering team development.

Ghaziabad Computer Centre, Ghaziabad, India

Trainee

August 2004 - May 2005

- Designed engaging product catalogues to enhance product visibility and customer interest, while supporting sales through phone and in-store customer interactions.
 - Supported computer hardware maintenance and troubleshooting to ensure smooth and efficient operations.
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AWARDS & ACHIEVEMENTS

- Delivered a global webinar on “UFT vs Selenium: Mobile Testing Strategy,” attracting 1,000+ participants.
 - Received Sustaining Edge Award (2012) for saving \$0.5 million with automation solutions at United Health Group.
 - Received Rock Star Award (2009) for achieving a 75% reduction in reporting generation effort at Sapient Corporation.
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EDUCATION & CERTIFICATIONS

- Bachelor of Technology in Computer Science & Engineering from Uttar Pradesh Technical University, 2000-2004.
- Experitest Seetest Certified Professional, License 201406231849.